

Wasatiyah framework for decision-making under complexity

Tariqullah Khan, MD Ventureethica, Toronto Canada

Feedback @ tariqullah.khan@gmail.com

Abstract

The moral epistemology of Islamic Economics seeks balance. In today's VUCA environment - characterized by volatility, uncertainty, complexity, and ambiguity - good decision-making has become more challenging than ever. Complex situations involve conflicts and tensions, which are embedded with inherent social, environmental, and financial costs, and demand more effective and just decision-making. Drawing inspiration from the concept of Wasatiyah (the middle path) in Islamic Economics, this paper presents a novel two-dimensional framework to navigate such complexities. This framework tackles the twin challenges of "noise" (information overload) and "tension" (competing objectives) by identifying and analyzing the two pivotal dimensions of any target circumstance. These dimensions are visualized using four distinct quadrants, facilitating a nuanced understanding of the decision-making landscape. A key feature of this framework is the use of dynamic diagnostics through Normalized Balanced Coordinates (NBCs). NBCs provide a unified measure for both dimensions, normalizing them on a -1 to +1 scale, enabling precise and comparable assessments across different situations. To guide decision-makers towards optimal outcomes, the framework incorporates a strategic approach utilizing Substitutions, Transformations, and Offsets (STOs). The versatility of this framework is demonstrated through its potential application across a wide range of sectors, from macroeconomic policies to sustainability initiatives.

Key words: Islamic Economics, Complexity, Normalized Balanced Coordinates, Decision theories, Wasatiyah

I Introduction

In today's VUCA environment - characterized by volatility, uncertainty, complexity, and ambiguity - good decision-making has become more challenging than ever (Bennett, N., & Lemoine, G. J. (2014) and Glenn, B. (2024)). The digital age, while providing unprecedented access to information, has also led to an explosion of both useful and misleading information, creating a fog of noise that clouds judgment. Social media amplifies this complexity by flooding decision-makers with unverified data, biased narratives, and conflicting viewpoints, often blurring the line between facts and misinformation. This deluge not only overwhelms cognitive capacities but also exacerbates tensions between competing objectives, making it difficult to identify priorities and align strategies. Navigating this fog requires not only clarity and focus but also a robust framework to filter out irrelevant noise, resolve inherent tensions, and create actionable pathways toward optimal outcomes.

Perceived complexity limits decision making. Consequently, undecidability emerges as a fundamental limit of formal reasoning, affecting decision-making in fields ranging from mathematics and computer science to social choice and game theory. Al-Suwailem, S. (2024) argues that undecidability is not merely an abstract computational issue but an intrinsic characteristic of complex systems, arising from paradoxes, self-referential classifications, and the impossibility of designing universally valid decision rules. This implies that traditional models relying on deterministic reasoning and logical completeness are insufficient for handling real-world complexity, necessitating alternative approaches to rational decision-making.

Complexity is the consequence of noise and tensions. Noise refers to irrelevant, misleading, or excessive information that obscures clarity and complicates the decision process. Noise can stem from information overload, conflicting data sources, or biases in data interpretation. For example, in economic policy, excessive attention to short-term market fluctuations can obscure long-term trends critical to sound decision-making. Noise disrupts the ability to focus on the core variables that matter most, making decisions more prone to errors or inefficiency. Reducing noise is essential to ensure decisions are based on accurate, concise, and relevant information.

Reducing noise often begins with narrowing down the dimensions at play. One useful approach is to identify classic trade-offs - those tension-laden, competing objectives that shape a decision. By mapping these opposing dimensions onto a conceptual framework, we can transform excessive, unfocused information into structured, meaningful insights. For instance, traditional polarities such as efficiency vs. equity, quality vs. quantity, growth vs. stability, or digitization vs. inclusion can help illuminate the core tension between optimizing resources and fostering social inclusivity.

Once the two pivotal dimensions are identified, the noise of sub-dimensions - previously scattered and overwhelming - can be organized as rubrics of the two dimensions, where each data point or piece of information is assessed for its relevance to either axis. This approach filters out irrelevant data, clarifies the relationships between variables, and creates a structured map of the decision space. By focusing on these two dimensions, decision-makers can turn unmanageable complexity into actionable intelligence, making it easier to diagnose the current state and devise pathways to improvement.

Tension arises when there are competing objectives or trade-offs in a decision. It reflects the inherent conflict between two desirable but opposing outcomes, such as efficiency vs. equity, growth vs. quality, or research freedom vs. ethical oversight etc. These tensions are common in complex systems where resources are limited, priorities diverge, and stakeholders' goals may conflict. Tension can stall progress, create inefficiencies, or lead to suboptimal compromises. Managing tension involves balancing these trade-offs to achieve the best possible outcome without disproportionately sacrificing one objective for another.

This balancing model draws inspiration from the concept of Wasatiyah (middle path, as emphasized in ancient scriptures and philosophical traditions. Rooted in the principle of moderation, it advocates for a balanced approach that avoids extremes, seeking harmony between competing forces. In the context of decision-making, this translates to navigating tensions such as efficiency versus equity or freedom versus oversight, etc., by finding a sustainable equilibrium that aligns with collective well-being. By integrating the wisdom of moderation, the model not only offers a pragmatic framework for resolving complex trade-offs but also imbues the process with a deeper ethical and philosophical foundation, encouraging decisions that are both just and inclusive. This timeless principle, reimagined through modern tools like normalized coordinates and dynamic transitions, serves as a guiding ethos for addressing complexity in an interconnected world.

Problem Statements

1.Noise

Too many apparently competing dimensions add to noise and eclipse the core and pivotal concerns of a complex situation. To clarify this dilemma, as an example, the complexity of modern sustainability frameworks, such as the 17 Sustainable Development Goals (SDGs) with 169 targets, has introduced excessive noise into decision-making and analysis (Spaiser, V., Ranganathan, S., Swain, R. B., & Sumpter, D. J. T. (2017). Whereas, in the SDGs, there is the fundamental tension between socio-economic development and the balance of natural systems United Nations. (2018). This proliferation of dimensions creates challenges such as:

- a) Information Overload: The sheer number of goals and targets overwhelms decision-makers, diluting focus on the fundamental tension and leading to inefficiency.
- b) Competing Priorities: In practice, the goals often conflict, particularly between human development goals (HDI) like poverty and hunger reduction, health and education on one hand versus the balance of natural system like climate stability and biodiversity conservation on the other hand – historically more developed countries have high ecological footprint United Nations (2024), (UNDP 2024), (GFN - Global Footprint Network (2024, 2024a).
- c) Fragmentation: The lack of a unified framework means goals are pursued in isolation, like hunger and poverty reduction in isolation of their impact on ecological balance leading to unintended trade-offs and systemic inefficiencies – without good quality of air and water poverty cannot be eliminated.

This excessive dimensionality makes it difficult to prioritize, measure, and achieve meaningful progress, often masking the core conflict between human development and the balance of natural systems.

2 Tensions

In traditional decision-making, two dimensions - for example, ethics vs profits, equity vs efficiency, full employment vs price stability (Hoover, K. D. (n.d.), quality vs growth (Angeli, S., Manning, T., Chung, J., & Howenstine, L. (2024, November 30) etc., are often treated as competing objectives, creating inherent tensions and trade-offs. The traditional approach assumes a zero-sum game: prioritizing one dimension necessitates compromising the other. For example, organizations and individuals struggle to balance short-term financial gains with long-term ethical obligations, often resulting in:

- a) Polarization: Ethical concerns are sidelined in pursuit of profitability, or profits are sacrificed to uphold ethics.
- b) Unsustainable Outcomes: Decisions driven by short-term gains exacerbate systemic externalities, such as environmental degradation or social inequalities and exclusion.
- c) Fragmented Metrics: Ethics and profits are measured independently, with no unified mechanism to evaluate their interdependencies, leaving decision-makers without clear guidance for achieving alignment.

This fragmented, tension-based approach leads to suboptimal results, undermining the potential for regenerative and sustainable solutions.

3. Dynamic transition

NBCs serve as aimed to diagnostic of a specific situation and its distance from the ideal win-win situation. Then we need continuous improvement within the ideal situation as well as transition strategies from transitioning from the sub-ideal situations to the ideal one. This dynamic optimization requires employing Substitutions (replacing harmful inputs with beneficial inputs), Transformations (systemic transformation), and Offsets (compensation for unavoidable harms) - STOs, to foster iterative improvements, ensuring that short-term goals align with long-term sustainability and resilience.

Objectives

To mitigate information overload by distilling complex scenarios into two pivotal dimensions (e.g., human development vs. ecological sustainability), thereby reducing noise and sharpening focus on core priorities.

To resolve systemic tensions through Normalized Balanced Coordinates (NBCs), a unified diagnostic tool that quantifies competing objectives (e.g., ethics vs. profits) on a harmonized scale (-1 to +1), enabling balanced assessments and iterative improvements.

To operationalize dynamic decision-making via Substitutions, Transformations, and Offsets (STOs), ensuring strategies evolve from imbalanced states (e.g., profit-driven excess) toward sustainable equilibrium, where ethical, economic, and ecological goals mutually thrive.

II Theoretical and methodological framework

Noise and Tension Reduction

Dimension Reduction

By focusing on two key dimensions, for example HDI and Ecological Footprint - we can significantly reduce noise, sharpen focus, and streamline decision-making. This two-dimensional framework offers several advantages:

- a) **Clarity and Simplicity:** Distilling the complex system into two primary dimensions highlights the central tension between human development and ecological balance, enabling clearer prioritization.
- b) **Integration of SDGs:** The 17 SDGs and their associated targets can be organized as rubrics within the respective two dimensions. The HDI dimension will include goals like poverty reduction, hunger alleviation, health, and education contributing to human development. Likewise, environmental subdimensions can be grouped under the balance of natural systems.
- c) **Balanced Perspective:** The ecological footprint dimension captures the impact of human activity on natural systems, ensuring environmental considerations are not overlooked in the pursuit of human development.
- d) **Enhanced Focus:** This framework highlights the trade-offs and synergies between HDI and ecological sustainability, enabling more strategic, balanced decision-making. Such an approach creates a single unified measure of the two dimensions that are traditionally seen as competing goals.

By adopting this simplified approach, decision-makers can avoid the pitfalls of fragmented priorities and information overload, focusing instead on achieving a harmonious balance between human progress and the preservation of natural systems. This reduction in noise ensures that the sustainability agenda becomes actionable, measurable, and effective.

Balancing trade-offs

A unified and balanced approach, inspired by the middle path, and operationalized through Normalized Balanced Coordinates (NBCs), offers a holistic solution by integrating - for example, Ethics versus Profits into a single, dynamic framework. Rather than treating ethics and profits as competing tensions, this approach harmonizes them to achieve synergy and mutual optimization with a single unified measure. Key features include:

- a) **Unified Measurement:** NBCs provide a normalized scale (-1 to +1) to evaluate as an example, ethics (Akhirah) and profits (Dunya) as interrelated rather than competing dimensions, enabling precise, balanced assessments.
- b) **Quadrant Analysis:** A Cartesian framework visualizes possible states (e.g., Quadrant 1: Positive ethics and profits), guiding decision-makers toward regenerative outcomes where both dimensions thrive.
- c) **Sustainable Solutions:** The approach eliminates trade-offs by focusing on integrative strategies, such as ethical investments, circular business models, and nature-positive solutions that benefit both society and the economy.

The unified and balanced approach reframes the tensions of the trade-off as opportunities for mutual alignment, creating a pathway for decision-making that is ethical, profitable, and sustainable. It exemplifies the essence of balancing by harmonizing the pursuit which are traditionally competing and are embedded with tensions.

Civilizational history as evolutionary data point

We propose Accumulated Epistemic Rationality (AER) as a conceptual framework to analyze humanity's collective capacity to generate, refine, and transmit knowledge across civilizational time that can be the basis of decision making. Unlike individualistic Epistemic Rationality (ER) - the ability of a single agent to form beliefs through evidence and experience as the basis of decision making (Steele, K., & Stefánsson, H. O. (2020) - AER reconceptualizes humanity itself as a unified meta-learner. This framework treats civilization as a transgenerational entity that accumulates, synthesizes, and optimizes knowledge through historical cycles of experimentation, crisis, and innovation.

Individual rationality is constrained by biological lifespans and localized contexts. AER, however, operates across millennia, integrating insights from ancient philosophical traditions, premodern institutions, and modern scientific paradigms. Historical figures e.g., Qarun/Korah as symbols of hubris in wealth accumulation (Qur'an (Surah Al-Qasas, Verses 76–82) and events become evolutionary data points in this process, illustrating recurring patterns in human behavior and societal adaptation.

Humanity's collective wisdom emerges not from isolated genius but from iterative refinement. For instance, the principle of Wasatiyah (balance/moderatism) transcends its Islamic origins to reflect a cross-cultural heuristic for avoiding extremes - a lesson distilled from centuries of societal overreach and collapse. Similarly, economic systems, governance models, and ethical codes encode hard-won insights into managing scarcity, power, and cooperation.

Modern manifestations

AER allows us to extract insights from civilizational narratives, treating them as data points for analysis. One such narrative is the story of Qarun (Korah) in the Qur'an (Surah Al-Qasas, Verses 76–82). It serves as a compelling case study, offering profound lessons on the ethical dimensions of wealth accumulation, lifestyle choices, and social responsibility. Qarun, had amassed vast wealth and riches that led him to an attitude of entitlement and arrogance, as he attributed his success solely to his own intelligence and efforts, disregarding the broader economic, social, and environmental systems that contributed to his prosperity. He remained fixated on material excesses and corruption, using his wealth as a symbol of status and power, attracting admiration from those who equated success with affluence. Ultimately, Qarun's failure to recognize the principles of balance led to his downfall, as his unchecked pursuit of wealth resulted in catastrophic consequences.

The story of Qarun underscores the necessity of balancing wealth with ethical and spiritual responsibility. His arrogance and hoarding of riches, while neglecting societal welfare and gratitude to divine provision, exemplify the dangers of materialism. The Islamic moral epistemology advocates for humility, moderation, and using wealth to support communities and environmental well-being, while also enjoying permissible pleasures. Prioritizing charity, ethical consumption, and long-term spiritual goals over excess fosters social harmony. Equally vital is

striking a balance between worldly ambitions and the Hereafter, ensuring material success does not eclipse spiritual growth or moral accountability.

Qarun's extravagance and corruption highlight the perils of unchecked wealth. His boastful display of riches fueled envy and inequality, contrasting with Islamic principles of simplicity, gratitude, and ethical resource management. Moderation demands avoiding waste, ostentation, and exploitative practices, instead promoting sustainable consumption and compassionate investments (Kamali, M. H. 2015, Biplob, Hossain, and Md Faruk Abdullah. 2021, Haneef, M. A., and Imon, R. A. (2024). Patience and a long-term perspective are essential to resist short-term indulgence, fostering self-discipline and systemic equity. Wealth must be wielded to create positive impacts, rejecting corruption and environmental harm in favor of justice and planetary stewardship.

Modern parallels to Qarun's excess manifest in extreme wealth inequality and ecological collapse, as seen in Piketty's $r > g$ (Piketty, T. (2014) model, where capital returns outpace economic growth, concentrating wealth and eroding political governance and social mobility (Wike, R., Fagan, M., Huang, C., Clancy, L., & Lippert, J. (2025, January 9). The top 1% disproportionately drive carbon emissions and resource depletion (Oxfam International. (2023, November 20), mirroring Qarun's hubris. The Wasatiyah (moderation) framework counters this by advocating wealth optimization instead of maximization and embedding economic, ecological social responsibilities. Addressing systemic risks requires aligning prosperity with effective offsets of negative externalities of markets, curbing both societal fragmentation and environmental degradation through sustainable, gratitude-driven practices that honor collective well-being over individual greed.

Wasatiyah (Balance) – Islamic moral epistemology framework

Qarun represents a mindset and an ideology in which ethics versus profits is an area of intense competition, tensions and trade-offs. Morale of the story teaches that a life of moderation - one that balances personal needs, social responsibilities, spiritual goals, and ethical values - is the path to lasting success and fulfillment. It warns against the extremes of greed, arrogance, and materialism while promoting a holistic and balanced approach to life, anchored in gratitude, simplicity, and a sense of responsibility to both society and the Creator. The epistemological foundations of these lessons are summarized in the supplication: "Our Lord, give us *in this world [that which is] good* and in the *Hereafter [that which is] good*, and protect us from the punishment of the Fire" (Surah Al-Baqarah, 2:201). Furthermore, "*And thus We have made you a just community that you will be witnesses over the people and the Messenger will be a witness over you*" (Surah Al-Baqarah, 2:143). *Here* the principle of moderation and justice is established, highlighting the role of the Muslim community as an example for others, embodying balance in faith, ethics, and actions. It emphasizes the responsibility of being witnesses to humanity, reflecting fairness, integrity, and adherence to divine guidance.

Balance of natural and social systems

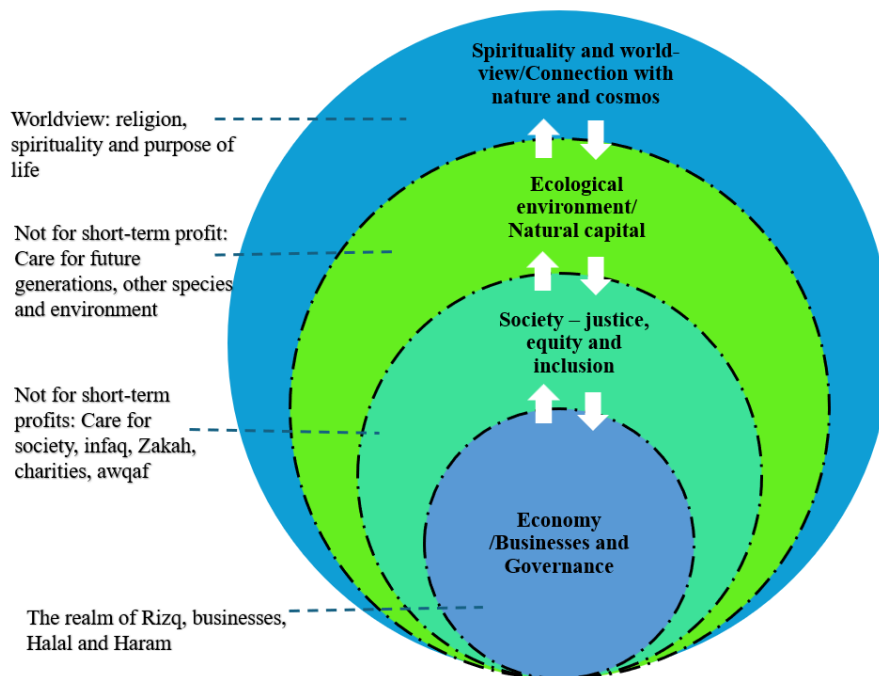
A central theme in Islamic Economics is to emphasize the holistic and human centric nature of the Islamic system. Siddiqi (1976) emphasizes the multidimensional development with a holistic perspective of the different dimensions of life.

The concept of *Meezan* (balance) in Islamic economics is rooted in an embedded epistemology that views knowledge and existence are interconnected. This worldview integrates economic,

spiritual, ethical, and ecological dimensions, recognizing humanity's role as stewards (*khalifah*) entrusted with maintaining equilibrium across natural, social, and economic systems. Grounded in the principle of *Tawhid*, it emphasizes harmony between material pursuits and spiritual purpose, rejecting reductionist approaches that isolate profit from planetary and societal health. The epistemology of *Meezan* positions balance as a divine mandate, where ecological cycles (water, carbon etc.), biodiversity, and resource stewardship reflect the balance of creation – “And the heaven He raised and imposed the balance. That you do not transgress within the balance. And establish weight in justice and do not make deficient the balance (Qur'an 55:7-9).

This holistic framework, as illustrated in the interconnected domains of economy, society, environment and spirituality Figure 1, asserts that true prosperity arises not from exploitation and maximization, but from aligning and optimizing human systems with the regenerative wisdom of nature. Knowledge here is relational - understanding humanity's embeddedness within cosmic order and ethical responsibility to sustain it.

Figure 1 Balanced, Holistic and Humane Development



Source: Drawn by author

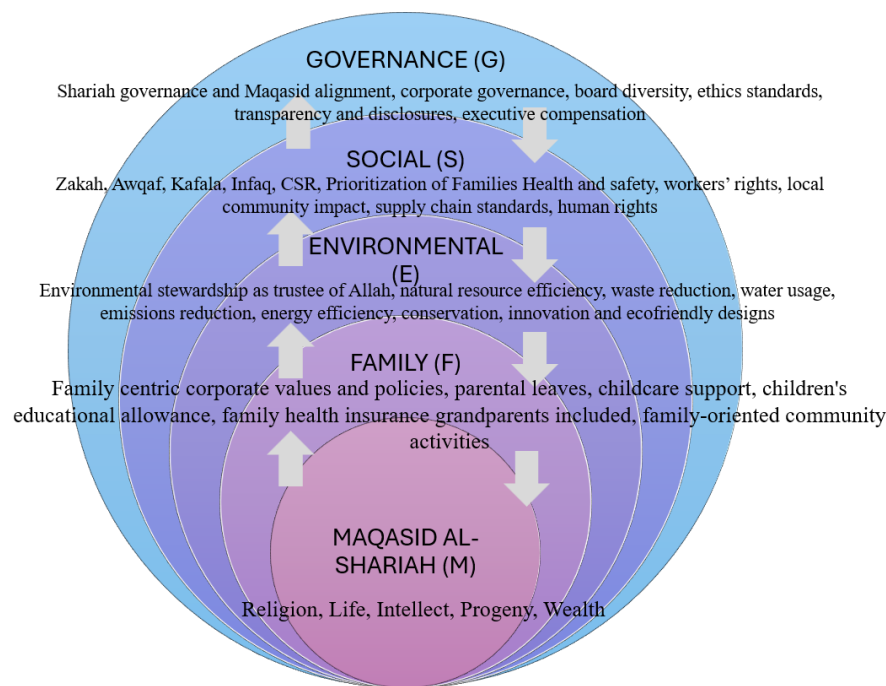
The balance of *Meezan* operationalizes this epistemology through systemic equilibrium, demanding justice (*adl*), equity, and circularity. It critiques linear, extractive models by advocating for circular economies that mirror natural cycles, minimizing waste and prioritizing resource renewal. Socially, it ties market activities to *infaq* (charity), *zakah*, and *awqaf* (endowments), ensuring wealth circulates to uplift communities rather than concentrate power. *Institutionally it strengthens these offsets of the market imperfections.* Ecologically, it binds human prosperity to the preservation of natural systems - air, water, and biodiversity - as sacred trusts (*amanah*). The embedded epistemology here rejects short-termism, framing sustainability as intergenerational equity and demographic harmony, where education, healthcare, and stable families cultivate resilience. By uniting spiritual reverence for nature with ethical governance, *Meezan* transforms balance from abstract ideal to actionable

praxis, ensuring economic systems thrive only when societies and ecosystems flourish in symbiotic unity.

Balance of objectives - Maqasid al Shariah

At present Google lists over 300 thousand works on the Maqasid, whereas Chapra (2008), Siddiqi (2009) are notably the first formal discussion of Maqasid al-Shariah with an urging note that Islamic Economics and Finance should be driven by the Maqasid. The first point to note here is that over 98% of the Google listing on Maqasid al-Shariah understand the Maqasid to be about protection and preservation of 1) Religion, 2) Life, 3) Intellect, 4) Progeny and 5) Wealth. However, Siddiqi (2009, 2014) doesn't restrict the Maqasid to these five; according to him the number of Maqasid shall depend on the society's needs. Historically these numbers have been increased but these have always remained in focus. For example, Tawheed (belief in Omnipotent, Omnipresent, Omnipotent God), Tazkiya (purification), teaching and empowering, respect for life and property, Meezan and justice, rights of different stakeholders etc., have always remained the Maqasid al-Shariah.

Figure 2 Proposed Maqasid and family aligned ESG (mf-ESG) framework



Source: Prepared by author

International interactions and economic relations are one of the main areas where Siddiqi calls for freedom of individual initiatives and incentives driven by the Maqasid. This suggestion of Siddiqi has far-reaching implications for Islamic Economics and Finance. There is a definite need for developing Maqasid scorecards and several researchers have also suggested such scorecards for guiding and measuring the performance of businesses based on the Maqasid. However, as we mentioned above the AAOIFI and IFSB standards were needed and rationalized to make Islamic financial services globally credible in all jurisdictions. With the same spirit and need the global sustainability standards that are driven by Environmental, Social and

Governance (ESG) considerations must be aligned with the Maqasid al-Shariah to obtain Maqasid and Families aligned ESGs (mf-ESG).

In recognition of the family as a core institution of human society, it is imperative that it be added as a separate and independent dimension within the Environmental, Social, and Governance (ESG) framework. Furthermore, for the ESG framework to be effectively applied within Muslim societies, it must be aligned with the principles of Maqasid al-Shariah. This alignment is visually and structurally represented in Figure 2, which redraws the ESG framework to place Maqasid al-Shariah and family dimensions at its core, thereby realigning the traditional ESG components to fit this broader and more integrated context.

The redesigned framework, referred to as the Maqasid and Family-aligned ESG (mf-ESG), is fully embedded in that it intricately interweaves the considerations of Maqasid, Family, and ESG. This integration ensures that each dimension reinforces the others, creating a robust system that profoundly supports the mission of balanced and humane development. Businesses that robustly apply this mf-ESG framework position themselves to contribute most effectively to this mission, as their operations and strategies become deeply rooted in fostering economic, social, and environmental well-being in alignment with Islamic ethical principles.

Wasatiyah Model

Figure 3 illustrates a framework for decision-making that balances Dunya (for-profit activities and considerations) and Akhirah (not-for-profit activities and considerations). The framework divides the decision space into four quadrants, defined by the normalized coordinates along the x-axis (representing Akhirah) and the y-axis (representing Dunya):

Quadrant I (Q I): Wasatiyah/Balanced Space

Coordinates: $X > 0$, $Y > 0$ (or $X = 0$, $Y = 0$ as the minimum threshold). This quadrant represents the optimal state of balance, where both for-profit and not-for-profit objectives are positively aligned. Activities within this space embody the principle of Wasatiyah (moderation), ensuring a harmonious integration of material success and spiritual or ethical considerations.

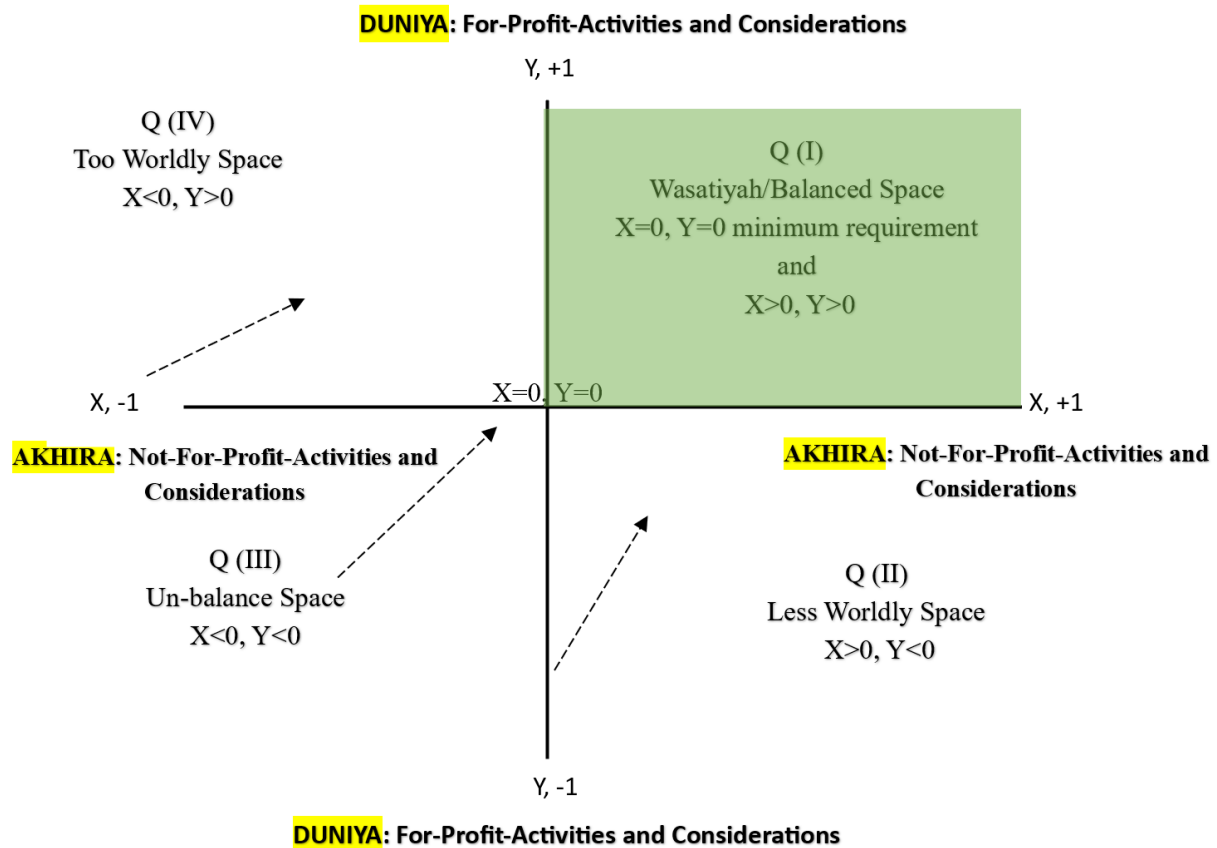
Quadrant II (Q II): Less Worldly Space

Coordinates: $X > 0$, $Y < 0$. This space represents activities that emphasize Akhirah (non-profit considerations) while sacrificing for-profit objectives. It reflects a disproportionate focus on ethical or spiritual goals at the expense of economic or material sustainability.

Quadrant III (Q III): Unbalanced Space

Coordinates: $X < 0$, $Y < 0$. This quadrant represents a state of failure in both dimensions, where neither for-profit nor not-for-profit considerations are adequately addressed. It is indicative of unsustainable and unethical activities, resulting in systemic inefficiencies and missed objectives.

Figure 3 - Wasatiyah framework of Islamic Economic Worldview



Source: drawn by author

Quadrant IV (Q IV): Too Worldly Space

Coordinates: $X < 0, Y > 0$. This space signifies a disproportionate focus on material or profit-driven activities (Dunya) at the expense of ethical or spiritual considerations (Akhirah). Such an approach may lead to negative externalities, including environmental degradation or social inequality.

The green-highlighted Quadrant I is the desired win-win state, representing a balanced, regenerative, and sustainable approach where both dimensions align to create positive, synergistic outcomes. This framework operationalizes the concept of Wasatiyah by integrating economic viability (Dunya) with ethical accountability (Akhirah). It provides decision-makers with a diagnostic tool to assess current activities, identify imbalances, and realign strategies toward achieving normalized positive outcomes in both dimensions.

Ethical Rubrics

It is often perceived difficult to develop a robust rubric for the ethical dimension. To further elaborate this important point, and to present a solution, in Table-1 we present a hypothetical example of 4 Halal restaurants. The total mf-ESG score are 100 and the Halal threshold scores are 20. The 4 Halal restaurants score differently on the different rubrics. Although all the 4

restaurants are Halal - Restaurant 1's mf-ESG scores are 45; Restaurant 2's mf-ESG scores are 60; Restaurant 3's mf-ESG scores are 80; and Restaurant 4's mf-ESG scores are 95.

Table 1 The Maqasid and Family aligned ESG (mf-ESG) of Halal Restaurants

<i>Mf-ESG Scores Total 100</i>	<i>mf-ESG Rubrics</i>	<i>Halal Restaurant 1</i>	<i>Halal Restaurant 2</i>	<i>Halal Restaurant 3</i>	<i>Halal Restaurant 4</i>
Maqasid Aligned ENVIRONMENTAL DIMENSION (mE) Total scores 20	Waste reduction Water usage Clean energy usage Biodiversity support Green supply chain Hyenine and usage of Plastic	2.5	10	20	20
Maqasid Aligned SOCIAL DIMENSION (S) Total scores 20	Zakah, Awqaf, Kafala, Infaq/CSR Families support Health and safety Workers' rights Local community impact Supply chain	2.5	10	10	20
Maqasid Aligned GOVERNANCE DIMENSION (G) Total scores 20	Maqasid alignment of Shariah governance Corporate governance and board diversity Ethics standards Transparency and disclosures Executive compensation	5	10	10	20
Maqasid Aligned FAMILY DIMENSION (F) Total scores 20	Family centric corporate values and policies, parental leaves, childcare support, children's educational allowance, family health insurance grandparents included, family- oriented community activities	15	10	20	15
Halal threshold scores 20		20	20	20	20
Total mf-ESG Scores of Restaurants		45	60	80	95

Source: Hypothetical Table, prepared by author for explanatory purposes

Note: Please refer to Figure 3 for the concept mf-ESG

Calculating NBCs

Table 2 NBCs for Halal Restaurants

Restaurants	mf-ESG Scores, from Table 1	Normalized X Coordinates	Normalized Y coordinates (assumed)	Normalized Balanced Coordinates (NBCs)	Quadrant Placement, reference Figure 3
Restaurant 1	45	-1	0.5	-1, 0.5	QVI, low on ethics
Restaurant 2	60	-0.4	0.5	-0.4,0.5	QVI, low on ethics
Restaurant 3	80	0.4	0.5	0.4, 0.5	QI, High on ethics
Restaurant 4	95	1	0.5	1. 0.5	QI, High on ethics

Source: NBCs calculation based on hypothetical Table 1

Note: Min-max normalization formula to scale values from -1 to +1

$X_{\min} = \min(\text{mf_ESG_scores})$, $X_{\max} = \max(\text{mf_ESG_scores})$

Normalized scores = $[2 * (x - X_{\min}) / (X_{\max} - X_{\min}) - 1]$ for x in mf_ESG_scores

Using the Min-max normalization formula to scale values from -1 to +1 we get the normalized balanced coordinates (NBCs) of Table 2 for the ethics dimension. Since the profits' rubric is not a challenge, we assume the profits' coordinates as 0.5 for all restaurants. Paradoxically, it is evident that Halal restaurants 1 and 2 fall into Quadrant 4 of Figure 3 scoring low on the ethical dimensions. The Wasatiyah model is helpful diagnostic to identify improvement areas.

III Diagnosis and Transformation

As shown in Table 2, a decision or system is diagnosed through Normalized Balanced Coordinates (NBCs) by mapping it within a two-dimensional Cartesian space, where X represents one fundamental dimension (e.g., sustainability, ethics) and Y represents another competing or complementary dimension (e.g., profitability, efficiency). NBCs normalize these dimensions onto a -1 to +1 scale, providing a quantitative diagnostic of the system's balance or imbalance. A positive NBC ($x > 0$, $y > 0$) signifies an optimal state (Quadrant 1), while negative coordinates indicate imbalances—such as profit-driven but unsustainable decisions (Q2), failing systems (Q3), or ethical but financially weak structures (Q4). Once the diagnosis is established, transformation occurs through Substitutions, Transformations, and Offsets (STOs), which are strategic interventions designed to shift decisions from imbalanced states toward equilibrium in Quadrant 1. Substitutions replace unsustainable or inefficient elements with balanced alternatives, Transformations restructure processes or governance for long-term viability, and Offsets compensate for unavoidable trade-offs by investing in complementary solutions. Together, NBCs and STOs form the two foundational pillars of the Wasatiyah framework, enabling a structured, iterative approach to achieving sustainable, ethical, and economically viable decision-making.

NBCs- further elaboration

Normalized Balanced Coordinates (NBCs) provide a unified measure of two dimensions (X and Y) in the Wasatiyah balancing framework. NBCs serve as a diagnostic tool that allows decision-makers to:

- Quantify the current state of a decision or system.
- Determine the balance or imbalance between competing priorities (e.g., ethics vs. profits, sustainability vs. economic growth).
- Track progress toward an ideal balanced state (Quadrant 1 in the Wasatiyah model).

NBCs transform qualitative trade-offs into a structured, measurable form, making it possible to assess whether a given decision is aligned with sustainability, ethical considerations, or optimal outcomes.

NBCs as a Unified Measure

NBCs normalize the two competing dimensions (X, Y) into a single quantitative scale (-1 to +1) so that:

- A positive NBC score (+1, +1) represents an ideal balanced state where both dimensions are optimized.
- A negative NBC score (-1, -1) indicates severe imbalance where neither objective is achieved.
- Intermediate values reflect partial progress or trade-offs between the two dimensions.

Each X, Y pair is plotted within the Cartesian system, and the corresponding NBC is computed to give a precise indication of the system's balance.

NBCs as a Diagnostic Tool for Decision-Making

NBCs help diagnose a system's position by identifying:

- Where the decision or system currently stands in the 4-quadrant space.
- How far the system is from the balanced optimal state (Quadrant 1: $x > 0$, $y > 0$).
- What types of strategic actions (Substitutions, Transformations, Offsets - STOs) are needed to move toward an optimal state.

Example: Diagnosing a Corporate Sustainability Strategy

- Dimension X: Environmental impact (negative to positive)
- Dimension Y: Economic performance (low to high)
- NBC score = (X, Y) = (-0.6, +0.3)
 - Interpretation: The company is prioritizing economic performance but has a negative environmental footprint.
 - Solution: Implement Substitutions (switch to sustainable supply chains) or Offsets (carbon credits, green investments) to improve X.

Thus, NBCs provide both a diagnostic and an improvement roadmap.

Constructing NBCs

To construct NBCs, we follow a structured approach:

Step 1: Define the Two Dimensions (X, Y)

The first step in constructing NBCs is identifying the two competing or balancing dimensions:

- X-axis: Represents one decision factor (e.g., ethics, sustainability, governance).
- Y-axis: Represents the competing or complementary factor (e.g., profits, efficiency, economic growth).

These dimensions must be quantifiable, even if indirectly.

Step 2: Normalize Each Dimension to [-1, +1]

Since different decision contexts use different scales (e.g., GDP growth, carbon emissions, ESG scores), each dimension must be normalized to ensure comparability.

The normalization formula is:

$$X_{normalized} = ((X_{actual} - X_{min}) / (X_{max} - X_{min})) \times 2 - 1$$

$$Y_{normalized} = ((Y_{actual} - Y_{min}) / (Y_{max} - Y_{min})) \times 2 - 1$$

where:

- X_{actual} , Y_{actual} are the measured values.
- X_{min} , X_{max} , Y_{min} , Y_{max} are the observed minimum and maximum values.

This transformation scales all values into a range of -1 to +1, ensuring that the NBCs are consistently interpretable.

Step 3: Compute the NBC Score

Once the X and Y dimensions are normalized, the NBC is simply represented as the coordinate pair (X, Y).

NBC = (X normalized, Y normalized)

This coordinate provides a precise, quantitative state of balance.

Step 4: Interpretation of NBCs

NBC Value (X, Y)	Interpretation	Implication
(+1, +1)	Balanced and ideal	Decision is aligned with both key objectives
(+1, -1)	Too profit-driven, ignoring ethical/sustainability concerns	Requires ethical/social/environmental offsets
(-1, +1)	Ethical/sustainable but financially unsustainable	Needs economic/financial stabilization
(-1, -1)	Failing in both dimensions	Requires fundamental restructuring

Step 5: Using NBCs for Continuous Improvement

- Once the NBC diagnoses a state, the next step is to move it toward balance.
- This is achieved using STOs:
 - Substitutions: Replace harmful components with beneficial alternatives.
 - Transformations: Change structural processes to improve sustainability.
 - Offsets: Compensate for negative externalities.

For example, a company with an NBC (-0.7, +0.2) (low sustainability, moderate economic success) might:

- Substitute fossil fuels with renewables (shifting X closer to 0 or positive).

- Transform operations to become circular.
- Offset their footprint by investing in community projects.

4. Example: Applying NBCs in Sustainability Finance

Use Case: Evaluating a Green Investment Portfolio

- X: Sustainability impact of investment (-1 = polluting industries, +1 = fully green)
- Y: Financial returns (-1 = negative ROI, +1 = high returns)
- NBC of Portfolio: (-0.3, +0.8)
 - Interpretation: The fund is highly profitable but only moderately sustainable.
 - Solution: Increase green finance allocation or use carbon offsets to improve X.

NBCs are a powerful diagnostic and decision tool because it quantifies complex trade-offs between two competing objectives and creates a unified measure of the two dimensions. It allows real-time monitoring and progress tracking in decision environments. NBCs provide a structured, objective way to evaluate balance and sustainability and guides decision-makers on how to transition toward a balanced state. Future research can expand NBCs into three-dimensional models and AI-driven decision-support tools.

STOs (Substitutions, Transformations, and Offsets)

Substitutions, Transformations, and Offsets (STOs) are dynamic mechanisms within the Wasatiyah balancing framework that enable decision-makers to continuously improve their position within the NBCs system. STOs serve two key purposes:

- Continuous Improvement in Quadrant 1 (Sustaining and Enhancing Balance)
- Transitioning from Quadrants 2, 3, and 4 to Quadrant 1 (Correcting Imbalances)

STOs as a Tool for Continuous Improvement in Quadrant 1

- Quadrant 1 (Q1) represents the optimal balance between two competing dimensions (e.g., sustainability vs. profitability, ethics vs. efficiency).
- Even within Q1, dynamic environments require continuous improvement to stay competitive and maintain balance.
- STOs serve as optimization levers to fine-tune and enhance performance in a sustainable and ethical manner.

STOs in Continuous Improvement (Q1 → Q1+)

STO Type	Definition	Example in Business and Policy	Impact
Substitutions	Replacing an existing component with a better alternative that enhances sustainability, efficiency, or ethics.	A company using renewable energy instead of fossil fuels.	Improves sustainability without reducing profitability.
Transformations	Redesigning or restructuring business models, governance, or policies for better	A circular business model that eliminates waste	Strengthens long-term resilience and market competitiveness.

	alignment with balanced outcomes.	and enhances resource efficiency.	
Offsets	Compensating for any remaining negative impacts by investing in positive externalities.	A company achieving carbon neutrality through tree-planting and offset investments.	Ensures net positive impact even in areas where full elimination of harm is impossible.

Even in the ideal quadrant, external shocks, market changes, and technological advances require continuous adaptation. STOs prevent stagnation and ensure that balanced states remain optimized over time.

STOs as a Tool for Transitioning from Quadrants 2, 3, and 4 to Quadrant 1

- Many systems, businesses, and policies start in suboptimal states (Quadrants 2, 3, or 4).
- STOs provide structured pathways for moving toward Quadrant 1 (optimal balance) by reducing trade-offs and addressing imbalances.

STO-Based Strategies for Quadrant Transitions

Quadrant	Problem Diagnosis	STO Strategy for Transition to Q1	Example
Quadrant 2 ($x < 0$, $y > 0$) (Economically successful but socially/environmentally harmful)	Prioritizes profit/growth while sacrificing ethics, sustainability, or governance.	Substitutions: Replace harmful practices with sustainable alternatives. Offsets: Invest in external ethical/sustainability programs to compensate for negative impacts.	Fossil-fuel-heavy industries transitioning to green energy while maintaining economic growth.
Quadrant 3 ($x < 0$, $y < 0$) (Failure in both economic and ethical/sustainability dimensions)	Systemic failure with poor outcomes in both dimensions, requiring fundamental changes.	Transformations: Overhaul governance, business models, or policy frameworks. Substitutions & Offsets: Gradual phasing out of harmful structures while adopting sustainable practices.	Failed economies implementing structural reforms to balance economic recovery with sustainability.
Quadrant 4 ($x > 0$, $y < 0$) (Socially/Ethically positive but economically unsustainable)	Strong ethical and sustainability commitments, but weak financial or operational viability.	Transformations: Introduce financially viable models that align with sustainability. Offsets: Leverage blended finance or subsidies to sustain ethical commitments.	NGOs or impact-driven businesses integrating profit-generating models (e.g., social enterprises) to

			enhance financial sustainability.
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STOs as a Dynamic Pathway: Incremental vs. Rapid Transitions

Incremental Transitions (Step-by-Step)

Some transitions require gradual improvements where Substitutions → Transformations → Offsets are applied in stages. For example, a traditional agriculture-based economy (Q3) slowly adopts regenerative farming (Q4) before fully integrating green technology and financial sustainability (Q1).

Rapid Transformations (Disruptive Changes)

Certain industries or policies must leapfrog directly into Q1 due to urgency or competitive pressure. For example, a government implementing strict carbon taxation to force rapid decarbonization in industries.

Summary: The Role of STOs in Dynamic Balancing

Scenario	Objective	STO Type Used
Q1 → Q1+ (Continuous Improvement)	Optimize balance further.	Substitutions, Transformations, Offsets applied incrementally.
Q2 → Q1 (Sustainability Deficit)	Correct economic-driven excesses by improving sustainability.	Substitutions & Offsets (e.g., sustainable supply chains, carbon credits).
Q3 → Q1 (Systemic Failure)	Structural overhaul required.	Transformations + Substitutions + Offsets combined in large-scale transition.
Q4 → Q1 (Economic Deficit)	Make ethical/sustainable models financially viable.	Transformations (e.g., social enterprises, government support).

Hence STOs are dynamic mechanisms that drive continuous improvement (Q1 → Q1+) and transitions from Quadrants 2, 3, and 4 to Q1.

Substitutions replace unsustainable practices, Transformations restructure decision frameworks, and Offsets compensate for negative impacts.

The Wasatiyah framework's flexibility allows STOs to be applied incrementally or disruptively, depending on the context.

Future research should explore the quantitative measurement of STO effectiveness and AI-driven decision-support models.

VI Generalization

Replacing the traditional trade-off between dimensions with a unified measure transforms decision-making by shifting the focus from conflict to resolution and balance. By introducing

normalized coordinates, each decision scenario can be represented as a single point on a two-dimensional plane, combining the pivotal dimensions into one coherent framework. For instance, in the tension between efficiency and equity, normalized coordinates quantify the balance, showing where the decision currently stands and where it needs to move. This unified measure replaces binary trade-offs with a spectrum of possibilities, enabling decision-makers to identify solutions that harmonize competing objectives rather than force a win-lose outcome. By visualizing the interplay of dimensions as a point on the plane, it becomes easier to foster consensus and compromise, ensuring that both dimensions are addressed in a balanced, measurable, and actionable manner. This approach transforms decision-making into a collaborative process of achieving optimal outcomes, reducing tension while promoting alignment and shared goals.

Partial List of Tensions

Here is a partial list of tensions where the framework can be applied to reduce noise, tensions and STOs for better decision making under complexity.

1. Macroeconomics

Price Stability vs. Full Employment (Balancing inflation control with job creation).

Real GDP vs. Cost of Finance (Balancing economic growth with affordable financing rates).

Fiscal Policy vs. Monetary Policy (Aligning fiscal spending with monetary policy objectives to ensure macroeconomic stability).

2. Banking and Finance

Efficiency vs. Stability (Pursuing operational efficiency while maintaining financial system resilience).

Ethics vs. Profit Maximization (Balancing profitability with ethical and sustainable practices).

Growth vs. Quality of Services (Scaling financial services rapidly without compromising service quality).

Cash flows vs. Sustainability (financial sustainability and ESG compliance).

3. Sustainability and ESG

Circularity vs. ESG Alignment (Balancing circular economy principles with broader ESG criteria).

Decarbonization vs. Economic Growth (Reducing carbon emissions while sustaining economic development).

Short-Term Profit vs. Long-Term Sustainability (Balancing immediate financial returns with regenerative practices).

Digital Growth vs. Environmental Impact (Expanding digital infrastructure while managing its carbon footprint).

4. Digitization and Technology

Inclusion vs. Innovation (Balancing accessibility and affordability with cutting-edge advancements).

Data Privacy vs. Utility (Protecting user privacy while leveraging data for innovation).

Automation vs. Employment (Increasing productivity through automation while managing job displacement risks).

Urban vs. Rural Digital Connectivity (Expanding digital infrastructure in urban areas vs. bridging rural digital divides).

Ethics vs. Technological Advancement (Ensuring ethical practices in AI and robotics without stifling innovation).

Open-Source vs. Proprietary Technology (Sharing innovations for collaboration vs. protecting intellectual property).

Speed of Adoption vs. Security (Deploying digital tools rapidly while ensuring robust cybersecurity).

Global Norms vs. Local Autonomy (Establishing global digital standards while respecting local laws and cultures).

Democratization vs. Power Concentration (Using digital tools to empower individuals while preventing monopolistic control).

5. Business Strategy

Quality vs. Growth (Maintaining high standards while scaling operations).

Short-Term Returns vs. Long-Term Goals (Prioritizing quarterly profits over strategic investments for the future).

Governance vs. Innovation (Balancing regulatory oversight with fostering disruptive innovation).

6. Decision-Making and Governance

Certainty vs. Welfare (Balancing confidence in outcomes with maximizing societal benefits).

Robustness vs. Optimization (Designing strategies resilient to uncertainty vs. highly efficient but less adaptable solutions).

Individual vs. Collective Outcomes (Balancing personal gain with societal welfare in strategic decisions).

Centralization vs. Decentralization (Centralized governance for efficiency vs. decentralized systems for autonomy).

Global Integration vs. Local Diversity (Aligning global strategies with local needs and cultural differences).

7. Public Utilities and Services

Efficiency vs. Equity (DoGE vs. DEI)

Public Education (Balancing cost-effective delivery with equitable access to quality education).

Public Health (Streamlining healthcare delivery while ensuring universal access and affordability).

Public Transport (Designing efficient transit systems while addressing inclusivity and affordability).

Public Housing (Optimizing housing solutions while ensuring affordable and accessible options for vulnerable populations).

8. Education and Research

Access vs. Quality (Expanding digital education while maintaining content excellence).

Research Freedom vs. Ethical Oversight (Allowing unrestricted exploration while ensuring ethical boundaries).

Traditional Knowledge vs. Digital Innovation (Preserving cultural heritage while digitizing and modernizing its use).

Generic Model

(Using Normalized Coordinates from -1 to 1)

In an environment characterized by volatility, uncertainty, complexity, and ambiguity (VUCA), effective decision-making requires tools that reduce irrelevant noise, manage tensions between competing objectives, and provide actionable strategies. This document presents a **generic two-dimensional framework** that organizes pivotal dimensions on a plane ranging from -1 to 1, enabling decision-makers to:

1. Filter out irrelevant or misleading information (noise).
2. Identify and measure tensions between two pivotal dimensions.
3. Employ Substitutions, Transformations, and Offsets (STOs) to navigate toward better outcomes.

Identifying Pivotal Dimensions

1. Dimension Set
 - Let us define two pivotal dimensions, D1 and D2, each capturing a critical yet potentially conflicting objective (e.g., Efficiency vs. Equity, Certainty vs. Welfare, etc.).
2. Examples of Dimensions
 - In Macroeconomics, D1 = Price Stability; D2 = Full Employment
 - In Sustainability, D1 = Decarbonization; D2 = Economic Growth
 - In Digitization, D1 = Data Privacy; D2 = Utility

These dimensions become the axes of our decision-making plane.

Normalization

1. Range

- Each dimension D1, D2 is normalized to lie within the interval $[-1, 1]$.
- D1 = -1 (worst possible state) up to +1 (best possible state).
- D2 = -1 (worst possible state) up to +1 (best possible state).

2. Data Mapping

- Gather raw data or metrics relevant to each dimension.
- Convert the raw metrics to a common scale of $[-1, 1]$.

This normalization ensures that both dimensions are **comparable** and easily visualized.

Noise

1. Definition

- Noise refers to irrelevant, misleading, or excessive information that obscures clarity in the decision process.

2. Relevance Function

- For each piece of information (data point) n in the total set N , define a relevance score $r(n)$ between 0 and 1.
- If $r(n) < \epsilon$ (a chosen threshold), the information is considered noise and filtered out.

3. Reduced Information Set

- By retaining only data points n for which $r(n) \geq \epsilon$, we reduce the information set to what directly informs D1 and D2.

Tension

1. Concept

- Tension arises when D1 and D2 are competing or negatively correlated. Improving one dimension might harm the other.

2. Tension Function $T(D1, D2)$

- $T(D1, D2) \geq 0$ is a measure of how difficult it is to improve both dimensions simultaneously.
- Example (Difference-Based):
 $T(D1, D2) = \max(0, \alpha - (D1 + D2))$,
where α is a constant threshold. If $D1 + D2$ is below α , tension is high.

3. Interpretation

- High $T(D1, D2)$ indicates a strong trade-off.
- Low $T(D1, D2)$ indicates possible synergy or easier alignment of objectives.

Quadrants

Because both $D1$ and $D2$ range from -1 to 1 , we have four quadrants in the decision plane:

1. Quadrant 1 (Q1):
 - $D1 \geq 0$ and $D2 \geq 0$.
 - Favorable on both dimensions; often the target or optimal zone.
2. Quadrant 2 (Q2):
 - $D1 \leq 0$ and $D2 \geq 0$.
 - One dimension is unfavorable ($D1$), one is favorable ($D2$).
3. Quadrant 3 (Q3):
 - $D1 \leq 0$ and $D2 \leq 0$.
 - Both dimensions are in an unfavorable state; typically, worst-case.
4. Quadrant 4 (Q4):
 - $D1 \geq 0$ and $D2 \leq 0$.
 - One dimension is favorable ($D1$), and the other is unfavorable ($D2$).

Single Unified Measure

Sometimes, decision-makers prefer to collapse both dimensions into a single scalar measure U ($D1, D2$):

1. Unified Performance Function
 - $U(D1, D2)$ lies in $[-1, 1]$.
 - Example (Weighted Sum): $U(D1, D2) = w1 * D1 + w2 * D2$, with $w1 + w2 = 1$.
 - Example (Minimum Operator): $U(D1, D2) = \min(D1, D2)$.
2. Interpretation
 - Higher $U(D1, D2)$ = better overall performance across both dimensions.
 - Lower $U(D1, D2)$ = suboptimal state or high tension.

STO Interventions

To move a given decision scenario ($D1, D2$) from a suboptimal quadrant (Q2, Q3, Q4) to the optimal quadrant (Q1), we employ Substitutions, Transformations, and Offsets.

1. Substitutions
 - Definition: Replace inefficient or harmful elements with more beneficial alternatives.
 - Mapping: $\sigma: S \rightarrow S$, where S is the 2D state space.

- Example: Replacing fossil fuels (low D2 for sustainability) with renewable energy sources (higher D2).

2. Transformations

- Definition: Redesign the underlying structure or governance to realign incentives and outcomes.
- Mapping: $\tau: S \times X \rightarrow S$, where X is a set of structural parameters.
 - Example: Overhauling organizational processes to improve both employee well-being (D2) and operational certainty (D1).

3. Offsets

- Definition: Compensate for unavoidable externalities or trade-offs.
- Mapping: $\omega: S \rightarrow S$.
 - Example: Carbon offsets that allow organizations to mitigate climate impacts while pursuing economic or technological objectives.

Putting It All Together

A typical decision-making workflow might look like this:

1. Identify and Normalize Dimensions
 - Pick the two pivotal dimensions and map them to $[-1, 1]$.
2. Filter Out Noise
 - Apply the relevance function $r(n)$ to remove data that does not directly inform D1 or D2.
3. Locate Current State
 - Plot the current scenario $S_0 = (D1, D2)$. Determine its quadrant.
4. Assess Tension
 - Calculate $T(S_0)$. If T is high, the trade-off between D1 and D2 is significant.
5. Develop STO Interventions
 - Propose Substitutions, Transformations, or Offsets to move S_0 toward a better quadrant.
6. Iterate
 - Move step by step, applying STOs until you reach an acceptable state or Quadrant 1.
 - Reassess the tension function T and/or the unified measure U to track improvement.

Example: Balancing Certainty (D1) and Welfare (D2) in decision theories

1. Normalization
 - $D1 = -1$: extremely high uncertainty; $D1 = +1$: high certainty.
 - $D2 = -1$: very low welfare; $D2 = +1$: very high welfare.
2. Current State:
 - Suppose $S0 = (-0.6, -0.2)$. This lies in Quadrant 3 (both dimensions unfavorable).
3. Tension Function:
 - Example : $T(D1, D2) = \max(0, 1 - (D1 + D2))$.
 - $T(-0.6, -0.2) = 1 - (-0.8) = 1.8$ (high tension).
4. Substitution
 - Replace unreliable forecasting methods with standardized, open-data methods to increase $D1$ from -0.6 to -0.2 .
5. Transformation
 - Restructure stakeholder engagement in policymaking to systematically improve welfare ($D2$) from -0.2 to 0.3 .
6. Offset
 - If additional welfare programs raise budget uncertainty, implement a fiscal offset (e.g., phased tax adjustments) to maintain overall stability (keeping $D1$ from dropping back down).
7. Result
 - New state might be $S1 = (0.2, 0.3)$. Tension is lower; both dimensions have improved. The system is closer to Quadrant 1, indicating better outcomes.

Summary

1. State Space: $S = [-1, 1] \times [-1, 1]$.
2. Noise Reduction: Define $r(n)$ for information n , filter out data with $r(n) < \epsilon$.
3. Tension: $T(D1, D2)$ measures trade-off severity between dimensions.
4. Quadrants: Q1 through Q4 categorize states based on $\text{sign}(D1)$, $\text{sign}(D2)$.
5. Unified Measure (Optional): $U(D1, D2)$ collapses both dimensions into one scalar for easier tracking.
6. STOs: Substitutions (σ), Transformations (τ), and Offsets (ω) navigate the 2D plane toward an optimal state.
7. Iteration: Decision-makers iteratively apply STOs to reduce tension and improve outcomes, ultimately aiming for Quadrant 1.

Mapping decision theories

The generic balancing model presented above is derived from the concept of Wasatiyah (the middle path) which is foundational in the epistemology of Islamic Economics. The key point to comprehend is to view human behaviour in a historical continuum. Qarun was a human being and the outcome of his behaviour - from ancient scriptures (as Kurah), can be reinterpreted through a modern analytical lens and enriched by decision theory frameworks. Epistemic rationality (ER) is an important concept in decision theories as it is about aligning one's beliefs with the evidence - how well one's beliefs correspond to reality and how effectively one can update them when one encounters new information (Steele, K., & Stefánsson, H. O. (2020)). As an accumulated epistemic rationality (AER), at its core learning from history, the principle of Wasatiyah advocates for moderation and equilibrium - values mirrored in this framework's focus on reducing tensions and "noise" in decision-making. By mapping decision theories onto a four-quadrant plane defined by Certainty (D1) and Welfare (D2) (or "Good Decision" in some formulations), the model operationalizes this balance in a structured and comparative way.

A key strength of the framework lies in its plasticity - numerous normalized coordinate points can coexist in each quadrant, reflecting diverse assumptions, contexts, and outcomes across various decision theories. In other words, the framework accommodates both high-level theoretical models and the many real-world variations within them.

To illustrate how different decision theories align with varying degrees of Certainty (D1) and Welfare (D2), we assign each theory to one of four quadrants, defined by whether D1 and D2 values are positive (≥ 0) or negative (≤ 0).

Quadrant 1 (High Certainty, High Welfare)

Theories: Wasatiyah Decision Theory - Rational Choice Theory (Idealized)

This is the "best-case" zone. Outcomes are both predictable (high certainty) and beneficial (high welfare). An idealized version of Rational Choice Theory, for example, 'invisible hand' (Smith 1776) operates within a society grounded in 'moral sentiments' (Smith 1759). Rational self-interest is tempered by empathy and social responsibility, assuming complete information and perfect logical processing, making it a natural fit for this quadrant. Here, decision-makers confidently and emphatically evaluate all alternatives and select the optimal one. A wise person learns from their mistakes (humanity in continuum learns lessons from Qarun/Kurah) and won't fall victim to the same egoistic trap twice. This aligns with the concept of accumulated epistemic rationality, where individuals update their beliefs and actions based on new evidence in their lives and learning from history and experiences to avoid repeating errors. In the context of rational choice theory, it suggests that a rational decision-maker will adjust their strategies after encountering a negative outcome to improve their chances of success in the future. The numerous possible NBCs within the quadrant will likely accommodate the various variants of rational choice (Becker, G. S. (1976) and game (Von Neumann, J., & Morgenstern, O. (1944) theories.

1. Learning from Historical Outcomes

Qarun's story serves as a timeless lesson in the importance of learning from past failures to avoid repeating destructive behaviors. His downfall, driven by arrogance and the unbalanced pursuit of wealth without regard for social responsibility or humility, underscores the dangers of neglecting ethical considerations in decision-making. In the human continuum, this tale

highlights how societies evolve by internalizing the lessons of such excesses, fostering a collective understanding of the need for balance between material success and moral accountability. By contextualizing Qarun in Quadrant 1, we recognize the value of historical learning as a foundation for high-certainty, high-welfare decision-making, where mistakes are not repeated, but instead inform better strategies for the future.

2. Alignment with Epistemic Rationality

Epistemic rationality involves the consistent updating of beliefs and actions considering new evidence and experiences, enabling individuals and societies to make more informed and balanced decisions. Qarun's refusal to heed warnings or adjust his behavior in response to evidence of his growing ego and misdeeds contrasts sharply with the principles of epistemic rationality. In Quadrant 1, decision-makers embody this rationality by refining their strategies after encountering negative outcomes, ensuring better alignment between their goals and reality. Contextualizing Qarun within this framework serves as a reminder that true rationality requires adaptability, humility, and a commitment to learning values that promote the harmony necessary for achieving optimal outcomes.

3. Relevance to Rational Choice Theory

Rational Choice Theory, particularly in its idealized form as mentioned above, assumes that agents act to maximize utility based on complete information and logical reasoning. Qarun's story demonstrates the failure of such rationality when driven by unchecked ego and the prioritization of self-interest over broader welfare. By linking Qarun's downfall to the absence of moderation, this narrative reinforces the importance of balanced decision-making, as modeled in Quadrant 1. Rational Choice Theory in this quadrant emphasizes the integration of material and ethical considerations, where decision-makers account for the collective good while pursuing personal objectives. Qarun's example thus serves as a counterpoint, illustrating the consequences of imbalance and the necessity of informed, rational choices that benefit both individuals and society.

Quadrant 2 (Low Certainty, High Welfare)

Theories: Satisficing Theories (Simon, H. A. (1956), Bayesian Approaches (Berger, J. O. (1985).

Despite imperfect or incomplete information (low certainty), decision-makers aim for outcomes that are still good or good enough (high welfare). Satisficing Theories recognize bounded rationality but still strive for acceptable solutions. Bayesian Approaches methodically update probabilities as information evolves, reflecting a structured path through uncertainty while seeking beneficial results.

Quadrant 3 (Low Certainty, Low Welfare)

Theories: Prospect Theory (Kahneman, D., & Tversky, A. (1979), Regret Theory - in certain scenarios, (Bell, D. E. (1982), Bounded Rationality (Simon, H. A. (1957).

Here, uncertainty is high, and outcomes tend to be poor or suboptimal. Prospect Theory (Kahneman and Tversky) highlights human biases - like loss aversion and reference dependence - leading to decisions that often deviate from strict rationality. Regret Theory (in its basic form) can also appear in this quadrant when the fear or experience of regret drives decisions away from optimal outcomes, further diminishing welfare. Likewise, Bounded

Rationality suggests limited cognitive resources and heuristic-driven decisions, raising the likelihood of error under uncertainty.

Quadrant 4 (High Certainty, Low Welfare)

Theories: Minimax Strategies (alternate game theories), Regret Theory (alternate case)

In this quadrant, outcomes are certain or predictable, yet they do not maximize welfare. Minimax Regret minimizes the maximum *regret* (difference between chosen action and the best possible action). Strategies reflect a strong preference for avoiding the worst-case scenario, often sacrificing potentially higher gains. Regret Theory can also manifest here when decision-makers knowingly choose a poor outcome (due to constraints or risk aversion) and later experience stronger regret precisely because the negative outcome was so predictable.

Analytical Justification for Mapping

Reduction of Complexity

Decision theories incorporate a range of assumptions that can be difficult to compare directly. Condensing their core characteristics into two normalized dimensions - Certainty and Welfare - provides a clearer, more tractable overview without oversimplifying the essential differences among them.

Unified Comparative Analysis

Placing these theories within a single, two-dimensional plane fosters more straightforward cross-disciplinary comparison. Economic, psychological, and strategic perspectives can thus be analyzed side by side, revealing both divergences and synergies.

Dynamic Plasticity

Each quadrant accommodates multiple coordinate points. This flexibility allows us to capture how variations in context, risk tolerance, or decision-maker preference can shift a given theory's manifestation from one region to another. Even the same theoretical framework may appear in different coordinates depending on the specific application or iteration.

Practical Application Through STOs

The framework facilitates improvement by introducing Substitutions, Transformations, and Offsets (STOs) - mechanisms that help shift decision-making trajectories toward more optimal quadrants, ideally Quadrant 1.

Substitutions: Replace limiting assumptions or rigid elements in a theory with more adaptable or context-specific alternatives. For example, substituting a purely Minimax perspective with a moderated, risk-tolerant approach can elevate welfare outcomes.

Transformations: Redesign the decision-making process. Incorporating AI or advanced data analytics into a bounded rationality framework can reduce uncertainty, moving decisions closer to higher-certainty, higher-welfare quadrants.

Offsets: Introduce compensatory mechanisms to counterbalance known pitfalls. In Prospect Theory, adopting systematic bias checks or "decision audits" can mitigate loss aversion effects, improving welfare even under uncertainty.

The Wasatiyah Connection

Wasatiyah underscores balance and harmony - a principle deeply woven into the model's design. Rather than forcing binary trade-offs, the model's plastic coordinate system lets decision-makers continuously adapt as contexts shift, reflecting an iterative practice of moderation. By aligning ancient wisdom with contemporary decision theory, the framework aspires to unify ethical considerations, practical strategies, and theoretical soundness into a cohesive guiding philosophy.

Important Considerations

Ideal vs. Real: These mappings simplify complex theories. In practice, real decisions often blend elements of multiple frameworks.

Dynamic Nature: The model allows us to “move” decisions via STOs (Substitutions, Transformations, Offsets)—e.g., from Q3 to Q1 by systematically reducing uncertainty and boosting welfare.

Context Matters: The precise placement of any decision theory depends on how Certainty and Welfare are measured and the specific context in which decisions are made.

By merging the Wasatiyah-inspired principle of balanced, harmonious decision-making with a robust two-dimensional mapping of theories, this framework provides both a theoretical lens and practical mechanisms (STOs) for improving decision outcomes. Whether one's focus is on rational optimization, managing uncertainty, or mitigating cognitive biases, the four-quadrant model clarifies how to approach decisions in ways that are mindful, adaptive, and ultimately more effective.

Oversimplification – An analysis of dimension reduction

One of the scepticisms about the framework is the risk of oversimplification of complex dimensions by reducing those to two. This is an acknowledged risk and to mitigate it we need comprehensive and robust rubrics of the sub-dimensions that are excluded.

Tables 1 and 2 demonstrate that a two-dimensional framework, when supported by robust rubrics, effectively reduces noise while retaining analytical depth. For instance: Table 1 (mf-ESG scores for Halal restaurants) organizes complex sustainability criteria (environmental, social, governance, family) into actionable rubrics. By aggregating these into a unified score, it filters irrelevant noise (e.g., redundant metrics) and highlights core tensions (e.g., ethics vs. profits). Table 2 (NBC calculations) translates these scores into normalized coordinates, enabling a clear quadrant-based diagnosis. This simplification allows decision-makers to prioritize improvements (e.g., moving from Quadrant IV to I via STOs) without losing sight of holistic goals.

Dimension reduction focuses on pivotal trade-offs (e.g., ethics vs. profits, HDI vs. ecological footprint etc.) and simplifies complex systems into diagnosable quadrants. The normalization process (-1 to +1 scale) ensures comparability across contexts. Hence, the risk of oversimplification is mitigated by the quality of rubrics. For example, the mf-ESG framework in Table 1 ensures that critical subdimensions (e.g., waste reduction, workers' rights) are not lost but aggregated into broader axes. This structured aggregation prevents fragmentation while preserving ethical and operational nuances.

While the two-dimensional model is sufficient for many scenarios, the framework’s design inherently supports expansion to three dimensions for richer analysis. Additional dimensions, however, increase computational and cognitive load and noise leading to increased “undecidability” or lower certainty or bad decisions.

The Wasatiyah framework’s two-dimensional design is not a rigid constraint but a modular foundation. Tables 1 and 2 prove that well-structured rubrics mitigate oversimplification risks, while the normalization process (NBCs) and improvement mechanisms (STOs) provide a scalable template for dimensionality expansion. Expand to three dimensions using the same rubric and normalization principles, enhancing the model’s diagnostic precision without sacrificing its ethical grounding is possible depending on the user preferences. This adaptability aligns with the framework’s philosophical roots in Wasatiyah (balance) and its methodological emphasis on iterative refinement. By allowing users to tailor dimensionality to their needs, the model remains both universally applicable and contextually precise.

V Comparing the Wasatiyah Framework with Established Decision Theories

The Wasatiyah framework for decision-making under complexity introduces a novel approach to balancing competing objectives in uncertain environments. Below, we compare its key principles, structure, and implications with established decision theories, including Rational Choice Theory, Bounded Rationality, Prospect Theory, Game Theory, Bayesian Decision Theory, and Behavioral Decision Theories.

1. Rational Choice Theory (RCT): Comparison

Feature	Wasatiyah Framework	Rational Choice Theory (RCT)
Decision Model	Balancing competing objectives using Normalized Balanced Coordinates (NBCs) and iterative Substitutions, Transformations, and Offsets (STOs)	Maximization of utility based on preferences and constraints
View on Trade-offs	Resolves trade-offs dynamically rather than treating them as fixed constraints	Trade-offs are necessary, decisions are made based on optimal utility
Rationality	Context-dependent rationality that integrates ethical, social, and economic dimensions	Assumes full rationality, perfect information, and logical consistency
Flexibility	Adapts to uncertainty and evolving preferences	Static model assuming fixed preferences
Criticism of RCT	Fails under real-world complexity, does not capture ethical and ecological considerations	Ignores ethical concerns, overestimates human cognitive ability

Key Takeaway: The Wasatiyah framework extends beyond rational choice by incorporating ethical, sustainability, and contextual factors into decision-making, making it more applicable to real-world complexity.

2. Bounded Rationality (Simon, 1957): Comparison

Feature	Wasatiyah Framework	Bounded Rationality (BR)
Decision Process	Uses NBCs to dynamically adjust decision pathways based on complexity	Decision-making is limited by cognitive constraints, heuristics are used
Optimization	Seeks a "balanced" rather than a maximized solution	Focuses on "satisficing" (finding a good-enough solution)
Handling Complexity	Reduces noise through dimension selection, making complexity more manageable	Uses heuristics to approximate optimal solutions within limits
View on Trade-offs	Trade-offs are minimized using STOs (iterative improvements over time)	Trade-offs exist, and the decision-maker accepts suboptimal solutions

Key Takeaway: Wasatiyah aligns with Bounded Rationality's realistic approach to complexity, but instead of satisficing, it seeks iterative improvements (STOs) to move toward balance rather than settling for a single decision.

3. Prospect Theory (Kahneman & Tversky, 1979): Comparison

Feature	Wasatiyah Framework	Prospect Theory
Treatment of Risk & Uncertainty	Uses NBCs to navigate uncertainty through gradual improvements	People overweight losses relative to gains, causing irrational choices
View on Trade-offs	Balanced decision-making aims to reduce tension dynamically	People make inconsistent choices depending on how a decision is framed
Biases and Heuristics	Encourages rational trade-off management by structuring objectives explicitly	Explains why people make errors but does not provide a corrective mechanism

Key Takeaway: Wasatiyah acknowledges human biases but provides structured tools (NBCs & STOs) to counteract them, whereas Prospect Theory primarily explains why people deviate from rationality but does not resolve those deviations.

4. Game Theory (Von Neumann & Morgenstern, 1944): Comparison

Feature	Wasatiyah Framework	Game Theory
Decision Focus	Focuses on individual and collective balance through ethical and sustainable outcomes	Focuses on strategic interaction between rational agents maximizing self-interest

View on Trade-offs	Seeks cooperative solutions to avoid a zero-sum outcome	Assumes decision-makers act in competition, often leading to zero-sum scenarios
Flexibility	Allows for dynamic adjustments in decision pathways based on real-world variables	Relies on fixed payoffs and strategies

Key Takeaway: Game Theory is useful for modeling strategic behavior but lacks a mechanism for balancing conflicting objectives in non-zero-sum contexts, which Wasatiyah addresses through STOs and NBCs.

5. Bayesian Decision Theory (Berger, 1985): Comparison

Feature	Wasatiyah Framework	Bayesian Decision Theory
Handling of Uncertainty	Uses a structured balance framework to manage uncertain environments	Uses probability updates to refine decision-making
Information Use	Incorporates ethical and long-term sustainability factors	Relies solely on probability and statistical inference
Trade-offs	Reduces tensions dynamically, rather than through probability-based expected utility calculations	Makes decisions based on expected utility but does not address ethical/sustainability concerns

Key Takeaway: Bayesian Decision Theory is statistically rigorous but does not integrate ethical or sustainability factors—Wasatiyah fills this gap by ensuring decisions align with long-term balance.

6. Behavioral Decision-Making (Thaler & Sunstein, 2008 - Nudge Theory): Comparison

Feature	Wasatiyah Framework	Behavioral Decision Theory (Nudge, Heuristics, Cognitive Biases)
Role of Biases	Provides a structured way to address biases by reducing noise and structuring choices dynamically	Explains and modifies biases through nudges and heuristics
Focus	Normative—seeks better decision-making outcomes based on balance	Descriptive—explains why people make irrational choices
Decision Mechanism	Provides iterative improvement strategies (STOs) to move toward balanced outcomes	Uses nudges and heuristics to guide behavior but does not explicitly quantify decision balance

Key Takeaway: Wasatiyah is normative (guiding decision-makers to better outcomes), whereas behavioral decision-making is descriptive (explaining why people make poor decisions). Wasatiyah incorporates structured improvement mechanisms (STOs), whereas nudges do not always ensure balance.

7. Wasatiyah Standing in Decision Theory

Theory	Strengths	Weaknesses Addressed by Wasatiyah
Rational Choice Theory	Optimal decision-making under fixed conditions	Fails in complex, uncertain, and ethical decision environments
Bounded Rationality	Recognizes cognitive limits	Accepts trade-offs rather than resolving tensions dynamically
Prospect Theory	Explains irrational decision-making	Does not provide corrective strategies for better decisions
Game Theory	Models strategic interactions	Lacks a cooperative, ethical, and sustainability-focused mechanism
Bayesian Decision Theory	Updates probabilities based on new information	Does not integrate ethical, environmental, and long-term sustainability considerations
Behavioral Decision-Making	Identifies cognitive biases and nudges behavior	Does not provide structured solutions to navigate decision complexity

In summary, Wasatiyah is a novel contribution to decision theory for several reasons:

- Balances competing objectives dynamically rather than relying on static trade-offs
- Integrates ethical, social, and environmental concerns with decision efficiency
- Provides structured improvement strategies (NBCs & STOs) to optimize outcomes iteratively
- Accommodates real-world complexity, uncertainty, and information overload

Thus, the Wasatiyah framework bridges the gap between rational optimization, behavioral realism, and ethical considerations, making it a practical and adaptable decision-making model for complex real-world scenarios.

IV Conclusions and Implications

Conclusions

The moral epistemology of Islamic Economics is based on the principle of Wasatiyah (balance and moderation) and the lessons to be drawn in perspective of the continuum of human experience. The Wasatiyah framework for decision-making under complexity offers a novel, structured approach to reducing noise and resolving tensions in decision environments characterized by volatility, uncertainty, complexity, and ambiguity (VUCA). Unlike traditional models that rely on binary trade-offs, the framework integrates Normalized Balanced Coordinates (NBCs) to establish a unified measure of pivotal dimensions such as ethics vs. profits, efficiency vs. equity, or sustainability vs. growth. This approach shifts decision-making from a zero-sum paradigm to one that seeks synergistic optimization through Substitutions, Transformations, and Offsets (STOs).

The framework's applicability across diverse domains - macroeconomics, sustainability, finance, technology, and governance - underscores its universality and robustness. By leveraging

historical epistemic rationality (AER) and grounding decision-making in the moderation principles of Wasatiyah, the model provides an ethical and adaptive alternative to conventional decision theories.

Given its conceptual novelty, empirical validation remains a critical next step. The model holds significant potential in academic research, policy analysis, and business strategy. However, as a complementary tool rather than a replacement for existing methodologies, its integration should be guided by critical evaluation and refinement through case studies and empirical research.

Research Implications

The Wasatiyah framework opens several research avenues across disciplines:

Dimensionality

One of the key strengths of the Wasatiyah balancing framework is its structural simplicity, which makes it applicable across various domains. However, depending on the specific use case, the framework can be flexibly adjusted to enhance its adaptability and analytical power. Future research should explore the expansion of the framework's dimensionality and quadrant flexibility to suit different decision environments.

Empirical Validation of the Framework

- Testing the effectiveness of NBCs in real-world decision-making.
- Applying the framework to case studies in macroeconomics, sustainability, finance, and business strategy.
- Developing quantitative models to measure improvements in decision outcomes when using STOs.

Advancing Decision Theory

- Exploring how the framework aligns with or improves upon existing decision theories (e.g., bounded rationality, prospect theory, satisficing, and game theory).
- Investigating the trade-off resolution capabilities of NBCs compared to classical optimization models.
- Assessing behavioral implications, such as whether decision-makers perceive less cognitive dissonance using a balanced rather than a trade-off approach.

Sustainability and Circular Economy Applications

- Using NBCs to redefine ESG and sustainability metrics.
- Exploring how Wasatiyah-based frameworks could guide corporate sustainability reporting and impact assessments.
- Investigating alternative economic models (e.g., Neo-Abrahamic economics) that integrate ethics and profits in a regenerative, non-exploitative way.

Islamic Economics and Finance

- Expanding research on Wasatiyah in financial decision-making.

- Examining how mf-ESG (Maqasid and Family-aligned ESG) can be integrated into sustainability finance standards.
- Developing Islamic financial instruments that incorporate STO-based decision processes to balance profit-seeking and social responsibility.

Pedagogical and Curriculum Innovation

- Introducing the Wasatiyah framework as a decision-theory paradigm in graduate programs.
- Developing teaching materials, case studies, and research themes that utilize NBCs.
- Integrating the Cartesian coordinate-based approach into business and policy analysis courses.

Policy Implications

The Wasatiyah framework has significant policy applications, particularly in sustainability, macroeconomic management, financial regulation, and digital governance.

Macroeconomic and Fiscal Policy

- Balancing price stability with employment growth: Policymakers can use NBCs to quantify trade-offs and employ STOs to navigate macroeconomic tensions (e.g., fiscal vs. monetary policy).
- Integrating ethical finance into public policy: The framework supports Wasatiyah-driven economic governance, ensuring fiscal policies align with social responsibility and intergenerational equity.

Sustainability and ESG Policy

- Redefining sustainability metrics: The framework provides a dynamic, non-binary ESG evaluation method that accounts for regenerative transitions rather than just incremental sustainability improvements.
- Harmonizing socio-economic development with ecological balance: Governments can use NBCs to assess policies that impact HDI vs. ecological footprint, optimizing interventions for human well-being without ecological overshoot.
- Guiding Circular Economy strategies: Policymakers can prioritize interventions that shift business models from Quadrant 3 (linear open-loop) to Quadrant 1 (circular-regenerative).

Islamic Finance and Ethical Investment Regulations

- Embedding Wasatiyah principles into financial regulations: Policies that promote blended finance mechanisms, such as Shariah-compliant Green Sukuk, can be aligned with NBC-based sustainability assessments.
- Enhancing fiduciary responsibilities in Islamic financial institutions by integrating mf-ESG criteria into regulatory requirements.
- Regulating speculative financial activities that diverge from Wasatiyah's balanced wealth ethics, countering excessive risk-taking.

Digital Economy and Governance

- Balancing data privacy vs. digital innovation: Policymakers can use NBCs to ensure privacy protection does not stifle AI advancements while preserving ethical standards.
- Regulating platform economies to ensure fair wealth distribution and avoid monopolistic tendencies.
- Ensuring AI governance aligns with balanced decision-making, applying NBCs to AI ethics frameworks.

Education and Public Sector Strategy

- Implementing Wasatiyah-based decision models in public policy schools.
- Encouraging public administration reforms that integrate NBCs to resolve trade-offs in education, healthcare, and public utilities.
- Using the model to guide ESG-aligned public investment strategies, ensuring balanced infrastructure development.

The Wasatiyah decision-making framework represents a paradigm shift in managing complexity. Its ability to quantify and balance competing objectives, combined with its theoretical foundations in Islamic moral epistemology and modern decision theory, makes it an innovative tool for both researchers and policymakers. Moving forward, its empirical testing, integration into governance models, and application in sustainability and financial regulations will be crucial to realizing its full potential as a transformative decision-support tool.

Declaration of generative AI and AI-assisted technologies in the writing process

The core intellectual content, including the framework, analysis, and arguments presented in this paper, were conceived and developed by the author. During the preparation of this work the author used ChatGPT 4o to improve language readability. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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